

CHAPTER Heat Conduction

HEAT TRANSFER THEORY

Heat transfer is an operation that occurs repeatedly in the food industry. Whether it is called cooking, baking, drying, sterilizing or freezing, heat transfer is part of the processing of almost every food. An understanding of the principles that govern heat transfer is essential to an understanding of food processing.

Heat transfer is a dynamic process in which heat is transferred spontaneously from one body to another cooler body. The rate of heat transfer depends upon the differences in temperature between the bodies, the greater the difference in temperature, the greater the rate of heat transfer.

Temperature difference between the source of heat and the receiver of heat is therefore the driving force in heat transfer. An increase in the temperature difference increases the driving force and therefore increases the rate of heat transfer. The heat passing from one body to another travels through some medium which in general offers resistance to the heat flow. Both these factors, the temperature difference and the resistance to heat flow, affect the rate of heat transfer. As with other rate processes, these factors are connected by the general equation:

$$\text{rate of transfer} = \text{driving force} / \text{resistance}$$

For heat transfer:

$$\text{rate of heat transfer} = \text{temperature difference} / \text{heat flow resistance of medium}$$

During processing, temperatures may change and therefore the rate of heat transfer will change. This is called *unsteady-state heat transfer*, in contrast to *steady-state heat transfer* when the temperatures do not change. An example of unsteady-state heat transfer is the heating and cooling of cans in a retort to sterilize the contents. Unsteady-state heat transfer is more complex since an additional variable, time, enters into the rate equations.

Heat can be transferred in three ways: by conduction, by radiation and by convection.

In conduction, the molecular energy is directly exchanged, from the hotter to the cooler regions, the molecules with greater energy communicating some of this energy to neighbouring molecules with less energy. An example of conduction is the heat transfer through the solid walls of a refrigerated store.

Radiation is the transfer of heat energy by electromagnetic waves, which transfer heat from one body to another, in the same way as electromagnetic light waves transfer light energy. An example of radiant heat transfer is when a foodstuff is passed below a bank of electric resistance heaters that are red-hot.

Convection is the transfer of heat by the movement of groups of molecules in a fluid. The groups of molecules may be moved by either density changes or by forced motion of the fluid. An example of convection heating is cooking in a jacketed pan: without a stirrer, density changes cause heat transfer by natural convection; with a stirrer, the convection is forced.

In general, heat is transferred in solids by conduction, in fluids by conduction and convection. Heat transfer by radiation occurs through open space, can often be neglected, and is most significant when temperature differences are substantial. In practice, the three types of heat transfer may occur together. For calculations it is often best to consider the mechanisms separately, and then to combine them where necessary.

HEAT CONDUCTION

In the case of heat conduction, the equation, rate = driving force/resistance, can be applied directly. The driving force is the temperature difference per unit length of heat-transfer path, also known as the temperature gradient. Instead of resistance to heat flow, its reciprocal called the conductance is used. This changes the form of the general equation to:

that is:

$$\begin{aligned} \text{rate of heat transfer} &= \text{driving force} \times \text{conductance}, \\ dQ/dt &= k A dT/dx \end{aligned} \quad (5.1)$$

where dQ/dt is the rate of heat transfer, the quantity of heat energy transferred per unit of time, A is the area of cross-section of the heat flow path, dT/dx is the temperature gradient, that is the rate of change of temperature per unit length of path, and k is the thermal conductivity of the medium. Notice the distinction between thermal conductance, which relates to the actual thickness of a given material (k/x) and thermal conductivity, which relates only to unit thickness.

The units of k , the thermal conductivity, can be found from eqn. (5.1) by transposing the terms

$$\begin{aligned} k &= dQ/dt \times 1/A \times 1/(dT/dx) \\ &= \text{Js}^{-1} \times \text{m}^{-2} \times 1/(\text{°C m}^{-1}) \\ &= \text{Jm}^{-1}\text{s}^{-1}\text{°C}^{-1} \end{aligned}$$

Equation (5.1) is known as the Fourier equation for heat conduction.

Note: Heat flows from a hotter to a colder body that is in the direction of the negative temperature gradient. Thus a minus sign should appear in the Fourier equation. However, in simple problems the direction of heat flow is obvious and the minus sign is considered to be confusing rather than helpful, so it has not been used.

Thermal Conductivity

On the basis of eqn. (5.1) thermal conductivities of materials can be measured. Thermal conductivity does change slightly with temperature, but in many applications it can be regarded as a constant for a given material. Thermal conductivities are given in Appendices 3,4,5,6, which give physical properties of many materials used in the food industry.

In general, metals have a high thermal conductivity, in the region $50\text{--}400 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Most foodstuffs contain a high proportion of water and as the thermal conductivity of water is about $0.7 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$ above 0°C , thermal conductivities of foods are in the range $0.6\text{--}0.7 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Ice has a substantially higher thermal conductivity than water, about $2.3 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$. The thermal conductivity of frozen foods is, therefore, higher than foods at normal temperatures.

Most dense non-metallic materials have thermal conductivities of $0.5\text{--}2 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Insulating materials, such as those used in walls of cold stores, approximate closely to the conductivity of gases as they are made from non-metallic materials enclosing small bubbles of gas or air. The conductivity of air is $0.024 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$ at 0°C , and insulating materials such as foamed plastics, cork and expanded rubber are in the range $0.03\text{--}0.06 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Some of the new in foamed plastic materials have thermal conductivities as low as $0.026 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

When using published tables of data, the units should be carefully checked. Mixed units, convenient for particular applications, are sometimes used and they may need to be converted.

Conduction through a Slab

If a slab of material, as shown in Fig. 5.1, has two faces at different temperatures T_1 and T_2 heat will flow from the face at the higher temperature T_1 to the other face at the lower temperature T_2 .

The rate of heat transfer is given by Fourier's equation:

$$dQ/dt = kA dT/dx$$

Under steady temperature conditions $dQ/dt = \text{constant}$, which may be called q :

$$\text{and so} \quad q = kA dT/dx$$

but dT/dx , the rate of change of temperature per unit length of path, is given by $(T_1 - T_2)/x$ where x is the thickness of the slab,

$$\begin{aligned} \text{so} \quad q &= kA(T_1 - T_2)/x \\ \text{or} \quad q &= kA \Delta T/x \\ &= (k/x) A \Delta T \end{aligned} \quad (5.2)$$

This may be regarded as the basic equation for simple heat conduction. It can be used to calculate the rate of heat transfer through a uniform wall if the temperature difference across it and the thermal conductivity of the wall material are known.

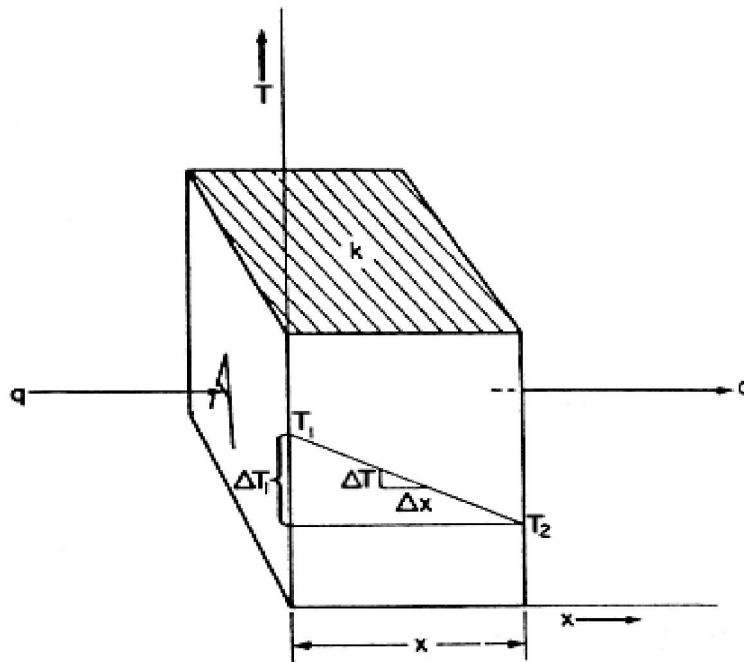


Figure 5.1. Heat conduction through a slab

EXAMPLE 5.1. Rate of heat transfer in cork

A cork slab 10cm thick has one face at -12°C and the other face at 21°C . If the mean thermal conductivity of cork in this temperature range is $0.042 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$, what is the rate of heat transfer through 1 m^2 of wall?

$$T_1 = 21^{\circ}\text{C} \quad T_2 = -12^{\circ}\text{C} \quad \Delta T = 33^{\circ}\text{C}$$

$$A = 1 \text{ m}^2 \quad k = 0.042 \text{ Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1} \quad x = 0.1 \text{ m}$$

$$\begin{aligned} q &= \frac{0.042}{0.1} \times 1 \times 33 \\ &= \underline{13.9 \text{ Js}^{-1}} \end{aligned}$$

Heat Conductances

In tables of properties of insulating materials, heat conductances are sometimes used instead of thermal conductivities. The heat conductance is the quantity of heat that will pass in unit time, through unit area of a specified thickness of material, under unit temperature difference. For a thickness x of material with a thermal conductivity of k in $\text{Jm}^{-1}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$, the conductance is $k/x = C$ and the units of conductance are $\text{Jm}^{-2}\text{s}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

$$\text{Heat conductance} = C = k/x.$$

Heat Conductances in Series

Frequently in heat conduction, heat passes through several consecutive layers of different materials. For example, in a cold store wall, heat might pass through brick, plaster, wood and cork.

In this case, eqn. (5.2) can be applied to each layer. This is illustrated in Fig. 5.2.

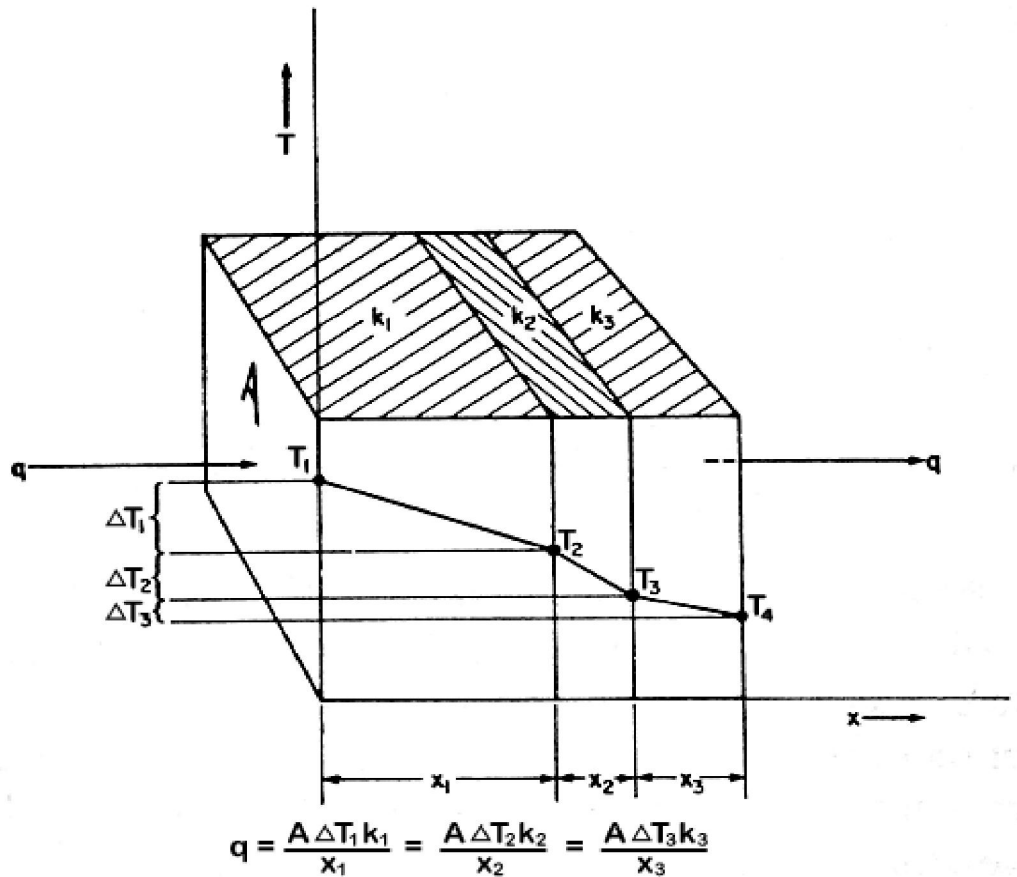


Figure 5.2 Heat conductances in series

In the steady state, the same quantity of heat per unit time must pass through each layer.

$$q = A_1 \Delta T_1 k_1 / x_1 = A_2 \Delta T_2 k_2 / x_2 = A_3 \Delta T_3 k_3 / x_3 = \dots\dots\dots$$

If the areas are the same,

$$A_1 = A_2 = A_3 = \dots\dots = A$$

$$q = A \Delta T_1 k_1 / x_1 = A \Delta T_2 k_2 / x_2 = A \Delta T_3 k_3 / x_3 = \dots\dots\dots$$

So $A\Delta T_1 = q(x_1/k_1)$ and $A\Delta T_2 = q(x_2/k_2)$ and $A\Delta T_3 = q(x_3/k_3)$

$$A\Delta T_1 + A\Delta T_2 + A\Delta T_3 + \dots = q(x_1/k_1) + q(x_2/k_2) + q(x_3/k_3) + \dots$$

$$A(\Delta T_1 + \Delta T_2 + \Delta T_3 + \dots) = q(x_1/k_1 + x_2/k_2 + x_3/k_3 + \dots)$$

The sum of the temperature differences over each layer is equal to the difference in temperature of the two outside surfaces of the complete system, i.e.

$$\Delta T_1 + \Delta T_2 + \Delta T_3 + \dots = \Delta T$$

and since k_1/x_1 is equal to the conductance of the material in the first layer, C_1 , and k_2/x_2 is equal to the conductance of the material in the second layer C_2 ,

$$\begin{aligned} x_1/k_1 + x_2/k_2 + x_3/k_3 + \dots &= 1/C_1 + 1/C_2 + 1/C_3 \dots\dots \\ &= 1/U \end{aligned}$$

where U = the overall conductance for the combined layers, in $\text{Jm}^{-2}\text{s}^{-1}\text{ }^\circ\text{C}^{-1}$.

Therefore $A \Delta T = q (1/U)$

And so $q = UA \Delta T$ (5.3)

This is of the same form as eqn (5.2) but extended to cover the composite slab. U is called the overall heat-transfer coefficient, as it can also include combinations involving the other methods of heat transfer – convection and radiation.

EXAMPLE 5.2. Heat transfer in cold store wall of brick, concrete and cork

A cold store has a wall comprising 11 cm of brick on the outside, then 7.5 cm of concrete and then 10cm of cork. The mean temperature within the store is maintained at -18°C and the mean temperature of the outside surface of the wall is 18°C .

Calculate the rate of heat transfer through 1m^2 of wall. The appropriate thermal conductivities are for brick, concrete and cork, respectively 0.69, 0.76 and $0.043 \text{ Jm}^{-1}\text{s}^{-1}\text{ }^\circ\text{C}^{-1}$.

Determine also the temperature at the interface between the concrete and the cork layers.

For brick $x_1/k_1 = 0.11/0.69 = 0.16$.

For concrete $x_2/k_2 = 0.075/0.76 = 0.10$.

For cork $x_3/k_3 = 0.10/0.043 = 2.33$

But
$$\begin{aligned} 1/U &= x_1/k_1 + x_2/k_2 + x_3/k_3 \\ &= 0.16 + 0.10 + 2.33 \\ &= 2.59 \end{aligned}$$

Therefore
$$\begin{aligned} U &= 0.38 \text{ Jm}^{-2}\text{s}^{-1}\text{ }^\circ\text{C}^{-1} \\ \Delta T &= 18 - (-18) = 36^\circ\text{C} \end{aligned}$$

$$A = 1\text{m}^2$$

$$\begin{aligned} q &= UA \Delta T \\ &= 0.38 \times 1 \times 36 \\ &= \underline{13.7\text{Js}^{-1}} \end{aligned}$$

Further, $q = A_3 \Delta T_3 k_3 / x_3$

and for the cork wall $A_3 = 1\text{ m}^2$, $x_3/k_3 = 2.33$ and $q = 13.7\text{Js}^{-1}$

Therefore
$$\begin{aligned} 13.7 &= 1 \times \Delta T_3 \times 1/2.33 \quad \text{from rearranging eqn. (5.2)} \\ \Delta T_3 &= 32^\circ\text{C}. \end{aligned}$$

But ΔT_3 is the difference between the temperature of the cork/concrete surface and the temperature of the cork surface inside the cold store.

Therefore $T_c - (-18) = 32$

where T_c is the temperature at the cork/concrete surface

and so $T_c = \underline{14^\circ\text{C}}.$

If ΔT_1 is the difference between the temperature of the brick/concrete surface T_b and the temperature of the external air

Then $13.7 = 1 \times \Delta T_1 \times 1/0.16 = 6.25 \Delta T_1$

Therefore $18 - T_b = \Delta T_1 = 13.7/6.25 = 2.2$

and so $T_b = \underline{15.8^\circ\text{C}}$

Working it through shows approximate interface temperatures: air/brick 18°C , brick/concrete 16°C , concrete/cork 14°C , and cork/air -18°C . This shows that almost all of the temperature difference occurs across the insulation (cork): the actual temperatures can be significant especially if they lie below the temperature at which the atmospheric air condenses, or freezes.

Heat Conductances in Parallel

Heat conductances in parallel have a sandwich construction at right angles to the direction of the heat transfer, but with heat conductances in parallel, the material surfaces are parallel to the direction of heat transfer and to each other. The heat is therefore passing through each material at the same time, instead of through one material and then the next. This is illustrated in Fig. 5.3.

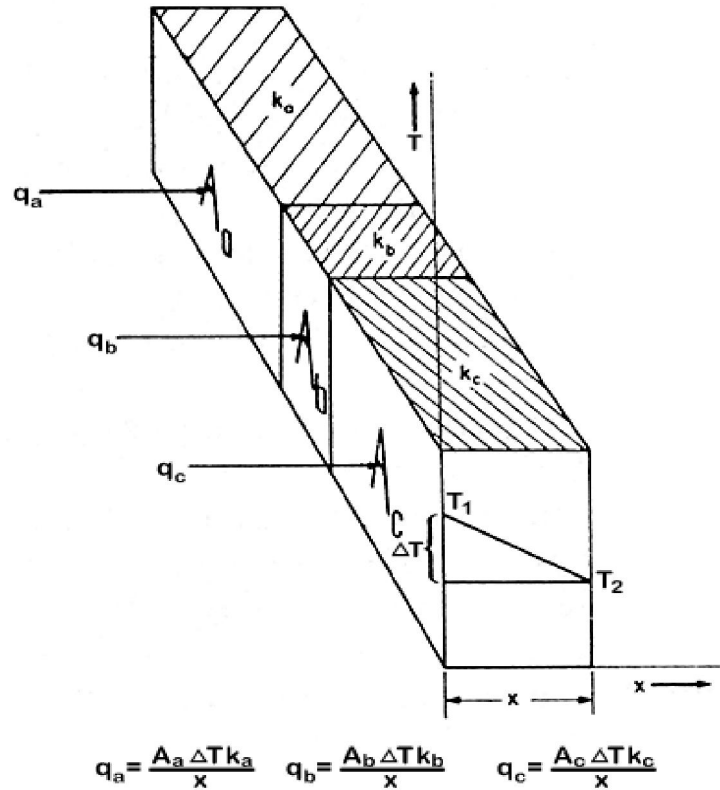


Figure 5.3. Heat conductances in parallel

An example is the insulated wall of a refrigerator or an oven, in which the walls are held together by bolts. The bolts are in parallel with the direction of the heat transfer through the wall: they carry most of the heat transferred and thus account for most of the losses.

EXAMPLE 5.3. Heat transfer in walls of a bakery oven

The wall of a bakery oven is built of insulating brick 10 cm thick and of thermal conductivity $0.22 \text{ Jm}^{-1}\text{s}^{-1}\text{°C}^{-1}$. Steel reinforcing members penetrate the brick, and their total area of cross-section represents 1% of the inside wall area of the oven.

If the thermal conductivity of the steel is $45 \text{ Jm}^{-1}\text{s}^{-1}\text{°C}^{-1}$, calculate (a) the relative proportions of the total heat transferred through the wall by the brick and by the steel and (b) the heat loss for each m^2 of oven wall if the inner side of the wall is at 230°C and the outer side is at 25°C .

Applying eqn. (5.1) $q = A \Delta T k/x$, we know that ΔT is the same for the bricks and for the steel. Also x , the thickness, is the same.

(a) Consider the loss through an area of 1 m^2 of wall (0.99 m^2 of brick, and 0.01 m^2 of steel)

$$\begin{aligned} \text{For brick } q_b &= A_b \Delta T k_b/x \\ &= \frac{0.99(230 - 25)0.22}{0.10} \\ &= 446 \text{ Js}^{-1} \end{aligned}$$

$$\begin{aligned}\text{For steel } q_s &= A_s \Delta T k_s / x \\ &= \frac{0.01(230 - 25)45}{0.10} \\ &= 923 \text{Js}^{-1}\end{aligned}$$

$$\text{Therefore } q_b / q_s = 0.48$$

(b) Total heat loss

$$\begin{aligned}q &= (q_b + q_s) \text{ per m}^2 \text{ of wall} \\ &= 446 + 923 \\ &= 1369 \text{Js}^{-1}\end{aligned}$$

$$\begin{aligned}\text{Therefore percentage of heat carried by steel} &= (923/1369) \times 100 \\ &= 67\%\end{aligned}$$

THIS TOPIC IS NOT THERE IN THE SYLLABUS SURFACE HEAT TRANSFER

Newton found, experimentally, that the rate of cooling of the surface of a solid, immersed in a colder fluid, was proportional to the difference between the temperature of the surface of the solid and the temperature of the cooling fluid. This is known as Newton's Law of Cooling, and it can be expressed by the equation, analogous to eqn. (5.2),

$$q = h_s A (T_a - T_s) \quad (5.4)$$

where h_s is called the surface heat-transfer coefficient, T_a is the temperature of the cooling fluid and T_s is the temperature at the surface of the solid. The surface heat-transfer coefficient can be regarded as the conductance of a hypothetical surface film of the cooling medium of thickness x_f such that $h_s = k_f / x_f$, where k_f is the thermal conductivity of the cooling medium.

Following on this reasoning, it may be seen that h_s can be considered as arising from the presence of another layer, this time at the surface, added to the case of the composite slab considered previously. The heat passes through the surface, then through the various elements of a composite slab and then it may pass through a further surface film. We can at once write the important equation:

$$\begin{aligned}q &= A \Delta T \left[\frac{1}{h_{s1}} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \dots + \frac{x_n}{k_n} + \frac{1}{h_{s2}} \right] \\ &= U A \Delta T\end{aligned} \quad (5.5)$$

where $1/U = (1/h_{s1}) + x_1/k_1 + x_2/k_2 + \dots + (1/h_{s2})$ and h_{s1}, h_{s2} are the surface coefficients on either side of the composite slab, $x_1, x_2, \dots$ are the thicknesses of the layers making up the slab, and $k_1, k_2, \dots$ are the conductivities of layers of

One dimensional Steady state heat conduction

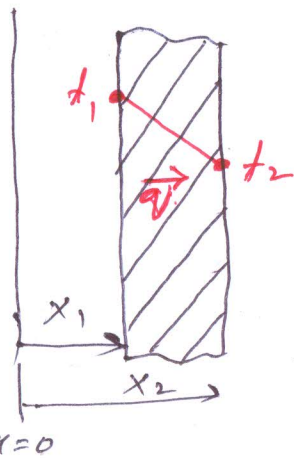


Fig: 1D steady state heat conduction through a plane wall.

$$q = \frac{t_1 - t_2}{R_k} = \frac{\delta}{KA}$$

For 1-D steady state heat conduction without heat generation,

$$\frac{\partial^2 t}{\partial x^2} = 0 \quad \text{--- (1)}$$

Integrating eqⁿ (1), we get $\frac{\partial t}{\partial x} = C_1$ --- (2)

and $t = C_1 x + C_2$ --- (3)

The Boundary conditions of the fig are at $x = x_1, t = t_1$ & at $x = x_2, t = t_2$. Substituting it to eqⁿ (3), which gives

$$t_1 = C_1 x_1 + C_2$$

$$t_2 = C_1 x_2 + C_2$$

The soln of these eq^s gives

$$C_1 = \frac{t_1 - t_2}{x_1 - x_2} = - \frac{t_1 - t_2}{x_2 - x_1}$$

$$\& C_2 = t_1 + \frac{t_1 - t_2}{x_2 - x_1} x_1$$

Now substituting values of C_1 & C_2 in eqⁿ (3) gives

$$t = - \frac{t_1 - t_2}{x_2 - x_1} x + \frac{t_1 - t_2}{x_2 - x_1} x_1 + t_1$$

or $\frac{t - t_1}{t_1 - t_2} = \frac{x_1 - x}{x_2 - x_1}$. This indicates a linear distribution of temp as shown in fig.

Now from eqⁿ (2) $\frac{\partial t}{\partial x} = C_1 = - \frac{t_1 - t_2}{x_2 - x_1}$.

From Fourier's law of heat equation. →

~~the~~ the rate of heat transfer is $q = -KA \frac{\partial t}{\partial x} = KA \frac{t_1 - t_2}{x_2 - x_1} = KA \frac{t_1 - t_2}{\delta}$

or $q = \frac{t_1 - t_2}{\delta / KA} = \frac{\Delta t}{R_k}$ (4) [Here $x_2 - x_1 = \delta$ thickness of plate]

Where A is the wall area perpendicular to the direction of 2/7
 heat flow & $R_K = \frac{\delta}{KA}$. K is thermal conductivity coefficient.
 & R_K thermal resistance of the wall.

Note The energy transferred by conduction & that heat transfer rate per unit area is proportional to normal temp. gradient.

$$\text{Thus } \frac{Q}{A} \propto \frac{dt}{dx} \quad \text{or} \quad Q = -KA \frac{dt}{dx}$$

Q is rate of heat transfer, $\frac{dt}{dx}$ is temp. gradient in dir. of heat flow. The minus sign is put so that the second law of thermodynamics is satisfied i.e. heat must flow down hill on the temp. scale. Q is (+)ive & $\frac{dt}{dx}$ is (-)ive. This is known as Fourier's law of heat conduction.

We can compare eq (4) with the Ohm's Law for an electrical conductor which is

$$I = \frac{E}{R} = \frac{V_1 - V_2}{R}. \text{ The electric current } I \text{ corresponds}$$

to the heat flow 'q', the el. potential E corresponds to the thermal potential & the electrical resistance corresponds to resistance R_K to the heat conduction.

Thus, the Fourier's eqⁿ of heat conduction is exactly analogous to Ohm's Law for an electrical conductor.

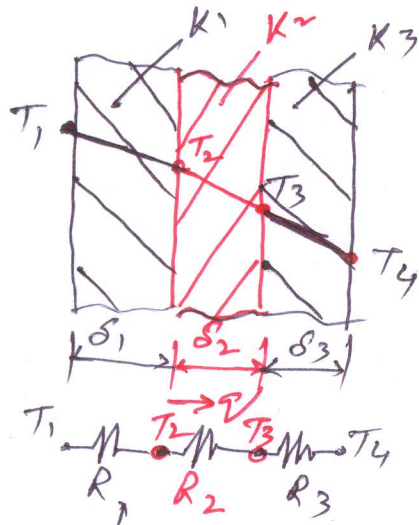
The reciprocal of R_K is known as thermal conductance of the wall.

$$t_1 \quad R = \frac{\delta}{KA} \quad t_2$$

Thermal Network.

$$K \Delta t \rightarrow$$

Thermal conduction in a composite wall



$$R_1 = \delta_1 / k_1 A$$

$$R_2 = \delta_2 / k_2 A$$

$$R_3 = \delta_3 / k_3 A$$

In the steady state condition, the heat flow q is the same for all the layers & is constant. The equations of heat transfer through these layers are

$$q = k_1 A \frac{T_1 - T_2}{\delta_1} \text{ for the first layer } - (i)$$

$$q = k_2 A \frac{T_2 - T_3}{\delta_2} \text{ for the 2nd layer } - (ii)$$

$$q = k_3 A \frac{T_3 - T_4}{\delta_3} \text{ for the 3rd layer } - (iii)$$

The temp differences across the layers, from the above equations, are

$$T_1 - T_2 = q \left(\frac{\delta_1}{k_1 A} \right) - (iv)$$

$$T_2 - T_3 = q \left(\frac{\delta_2}{k_2 A} \right) - (v)$$

$$T_3 - T_4 = q \left(\frac{\delta_3}{k_3 A} \right) - (vi)$$

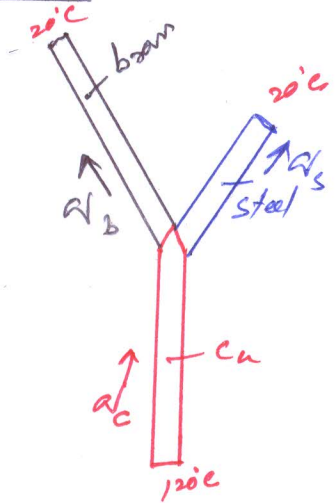
Adding the above equations, we get

$$T_1 - T_4 = q \left(\frac{\delta_1}{k_1 A} + \frac{\delta_2}{k_2 A} + \frac{\delta_3}{k_3 A} \right) - (vii)$$

$$q = \frac{(T_1 - T_4)}{\left(\frac{\delta_1}{k_1 A} + \frac{\delta_2}{k_2 A} + \frac{\delta_3}{k_3 A} \right)} = \frac{(T_1 - T_4)}{(R_1 + R_2 + R_3)}$$

Example on heat conduction (1D, steady state)

Rods of Copper, brass & steel are welded together to form a Y shaped fig as shown. The cross-section area of each rod is 400 mm^2 . The end of Cu rod is maintained at 120°C & the ends of brass & steel rods at 20°C . The lengths of rods are: Cu 768 mm, brass 172 mm, & steel 45 mm. Find



(a) What is the temp of the junction point?

(b) What is the heat flowing in Cu rod?

The thermal conductivity of Cu, brass & steel are 384, 86 & 45 W/mK respectively. The rods are insulated from outside.

Soln: Heat balance at junctions gives:

Heat inflow at the junction thru Cu rod, q_c = heat out at the junction thru brass & steel rods ($q_b + q_s$)

Where, from Fourier's eqⁿ:

$$q_c = KA \frac{\Delta t}{L} = 384 \times \frac{400}{(1000)^2} \times \frac{120 - t}{0.768} = 0.2 \times (120 - t)$$

$$q_b = KA \frac{\Delta t}{L} = 86 \times \frac{400}{(1000)^2} \times \frac{t - 20}{0.172} = 0.2 \times (t - 20)$$

$$q_s = KA \frac{\Delta t}{L} = 45 \times \frac{400}{(1000)^2} \times \frac{t - 20}{0.045} = 0.4 \times (t - 20)$$

Substituting in heat balance eqⁿ gives

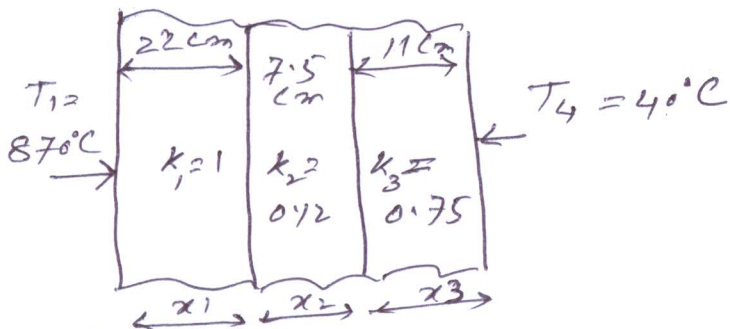
$$0.2 \times (120 - t) = 0.2 \times (t - 20) + 0.4 \times (t - 20)$$

$$\text{or } \boxed{t = 45^\circ\text{C}} \text{ Ans.}$$

Thus given $\boxed{q_c = 0.2 \times (120 - 45) = 15 \text{ W}}$ Ans.

Example y 1-D heat conduction through composite walls. 5/7

A furnace wall is made up of three layers, one of fire brick, another of insulating brick & one of red brick. The inner and outer surfaces are at 870°C & 40°C respectively. The respective co-efficient of thermal conductivities of layers are 1.0, 0.12 and $0.75 \text{ W/m}\cdot\text{K}$ & the thicknesses are 22 cm, 7.5 cm & 11 cm as shown in fig. Assuming close bonding of the layer at their interfaces. Find the rate of heat loss per sq meter per hour & the interface temperatures (T_2, T_3).



Soln

$$\frac{Q}{A} = \frac{T_1 - T_4}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}} = \frac{870 - 40}{\frac{22}{100 \times 1} + \frac{7.5}{100 \times 0.12} + \frac{11}{100 \times 0.75}} = 835 \text{ J/s} = \text{W}$$

$$= 835 \times 3.6 = 3006 \text{ kJ/hr}$$

also $\frac{Q}{A} = \frac{(T_1 - T_2)k_1}{x_1} = \frac{(T_2 - T_3)k_2}{x_2} = \frac{(T_3 - T_4)k_3}{x_3}$

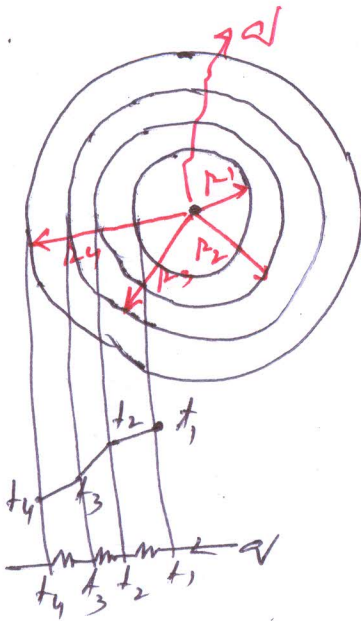
$$\therefore 835 = \frac{(T_1 - T_2)k_1}{x_1} \therefore T_2 = T_1 - 835 \times \frac{x_1}{k_1} = 870 - 835 \times \frac{22}{100 \times 1} = 686.5^{\circ}\text{C}$$

Similarly, $T_3 = T_2 - 835 \times \frac{x_2}{k_2} = 686.5 - 835 \times \frac{7.5}{0.12 \times 100} = 163.5^{\circ}\text{C}$

$$\therefore \boxed{T_1 = 870^{\circ}\text{C}, T_2 = 686.5^{\circ}\text{C}, T_3 = 163.5^{\circ}\text{C} \text{ \& } T_4 = 40^{\circ}\text{C}}$$

Ans.

1D steady state radial heat conduction thru Composite walls



The composite cylinder made of layers of different Materials. as shown.

$$q = \frac{t_1 - t_2}{\frac{1}{k_1 2\pi L} \ln \frac{r_2}{r_1}} \quad \text{--- (1)}$$

$$\text{also } q = \frac{t_2 - t_3}{\frac{1}{k_2 2\pi L} \ln \frac{r_3}{r_2}}$$

$$\text{also } q = \frac{t_3 - t_4}{\frac{1}{k_3 2\pi L} \ln \frac{r_4}{r_3}}$$

L is length of composite walls $\perp$ to the paper.

$$\therefore (t_1 - t_2) = \frac{q}{2\pi k_1 L} \ln \frac{r_2}{r_1}$$

$$(t_2 - t_3) = \frac{q}{2\pi k_2 L} \ln \frac{r_3}{r_2}$$

$$\& (t_3 - t_4) = \frac{q}{2\pi k_3 L} \ln \frac{r_4}{r_3}$$

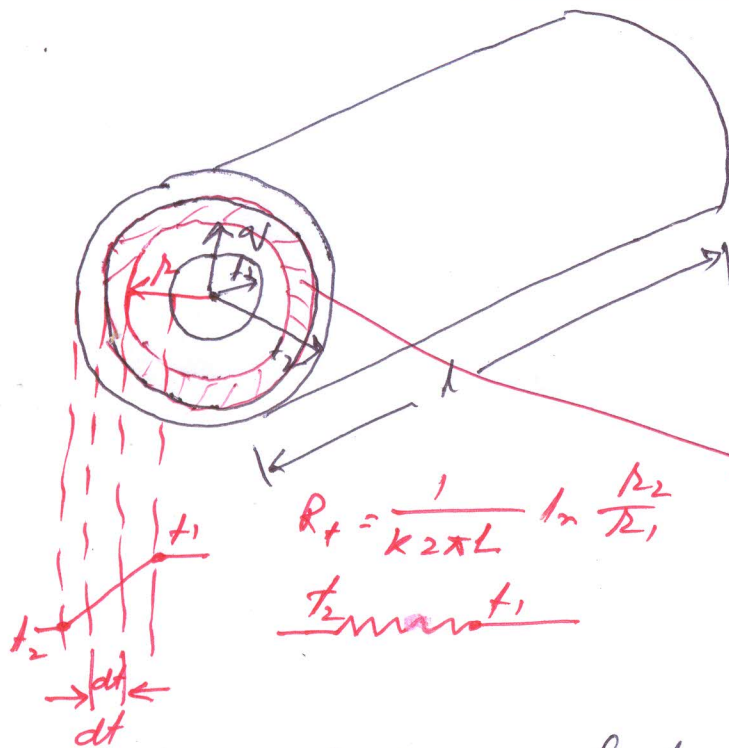
$$\therefore (t_1 - t_4) = \frac{q}{2\pi L} \left[\frac{1}{k_1} \ln \frac{r_2}{r_1} + \frac{1}{k_2} \ln \frac{r_3}{r_2} + \frac{1}{k_3} \ln \frac{r_4}{r_3} \right]$$

$$q = \frac{(t_1 - t_4)}{\frac{1}{2\pi k_1 L} \ln \frac{r_2}{r_1} + \frac{1}{2\pi k_2 L} \ln \frac{r_3}{r_2} + \frac{1}{2\pi k_3 L} \ln \frac{r_4}{r_3}}$$

$$q = \frac{(t_1 - t_4)}{R_{t1} + R_{t2} + R_{t3}}$$

1D Steady state ^{Radial} conduction thru a hollow cylinder

77



r_1 = internal radius

r_2 = external "

t_1 = internal surface temp.

t_2 = external surface temp.

$$R_t = \frac{1}{k 2\pi L} \ln \frac{r_2}{r_1}$$

$$t_2 \text{ max } t_1$$

Strip of dr

By Fourier's Law of heat conduction:

$$Q = -KA \frac{dt}{dx} = -k 2\pi L \frac{dt}{dr} \quad (1)$$

$$dt = \frac{-Q}{k 2\pi L} \frac{dr}{r} \quad (2)$$

Integrating between limits t_1 & t_2

$$(t_1 - t_2) = \frac{Q}{k 2\pi L} [\ln r_2 - \ln r_1] \quad (3)$$

$$= \frac{Q}{k 2\pi L} \ln \frac{r_2}{r_1}$$

$$\therefore Q = \frac{k 2\pi L (t_1 - t_2)}{\ln \frac{r_2}{r_1}} = \frac{t_1 - t_2}{\frac{1}{k 2\pi L} \ln \frac{r_2}{r_1}} = \frac{t_1 - t_2}{R_t} \quad (4)$$

$$\therefore R_t = \frac{1}{k 2\pi L} \ln \frac{r_2}{r_1}$$

Why is the temperature gradient dT/dx not continuous? (MP HT.6)

Why is ΔT the same for the two elements in a parallel thermal circuit? Doesn't the relative area of the bolt to the wood matter? (MP HT.7)

2.3 Steady Quasi-One-Dimensional Heat Flow in Non-Planar Geometry

The quasi one-dimensional equation that has been developed can also be applied to non-planar geometries. An important case is a cylindrical shell, a geometry often encountered in situations where fluids are pumped and heat is transferred. The configuration is shown in Figure 2.9.

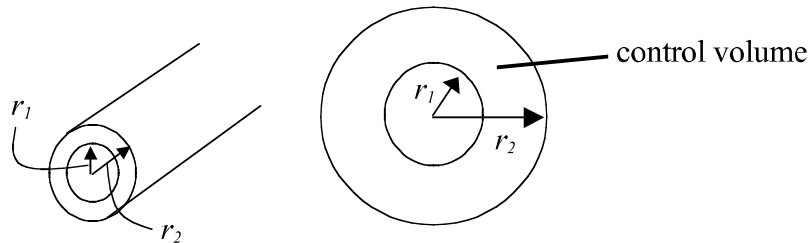


Figure 2.9: Cylindrical shell geometry notation

For a steady axisymmetric configuration, the temperature depends only on a single coordinate (r) and Equation (2.12b) can be written as

$$k \frac{d}{dr} \left(A(r) \frac{dT}{dr} \right) = 0 \quad (2.23)$$

or, since $A = 2\pi r$,

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0. \quad (2.24)$$

The steady-flow energy equation (no flow, no work) tells us that $\dot{Q}_{in} = \dot{Q}_{out}$ or

$$\frac{d\dot{Q}}{dr} = 0 \quad (2.25)$$

The heat transfer rate per unit length is given by

$$\dot{Q} = -k \cdot 2\pi r \frac{dT}{dr}.$$

Equation (2.24) is a second order differential equation for T . Integrating this equation once gives

$$r \frac{dT}{dr} = a . \quad (2.26)$$

where a is a constant of integration. Equation (2.26) can be written as

$$dT = a \frac{dr}{r} \quad (2.27)$$

where both sides of equation (2.27) are exact differentials. It is useful to cast this equation in terms of a dimensionless normalized spatial variable so we can deal with quantities of order unity. To do this, divide through by the inner radius, r_1

$$dT = a \frac{d(r/r_1)}{(r/r_1)} \quad (2.28)$$

Integrating (2.28) yields

$$T = a \ln\left(\frac{r}{r_1}\right) + b . \quad (2.29)$$

To find the constants of integration a and b , boundary conditions are needed. These will be taken to be known temperatures T_1 and T_2 at r_1 and r_2 respectively. Applying $T = T_1$ at $r = r_1$ gives $T_1 = b$. Applying $T = T_2$ at $r = r_2$ yields

$$T_2 = a \ln \frac{r_2}{r_1} + T_1 ,$$

or

$$a = \frac{T_2 - T_1}{\ln(r_2 / r_1)} .$$

The temperature distribution is thus

$$T = (T_2 - T_1) \frac{\ln(r/r_1)}{\ln(r_2/r_1)} + T_1 . \quad (2.30)$$

As said, it is generally useful to put expressions such as (2.30) into non-dimensional and normalized form so that we can deal with numbers of order unity (this also helps in checking whether results are consistent). If convenient, having an answer that goes to zero at one limit is also useful from the perspective of ensuring the answer makes sense. Equation (2.30) can be put in non-dimensional form as

$$\frac{T - T_1}{T_2 - T_1} = \frac{\ln(r/r_1)}{\ln(r_2/r_1)}. \quad (2.31)$$

The heat transfer rate, $\dot{Q}$, is given by

$$\dot{Q} = -kA \frac{dT}{dr} = -2\pi r_1 k \frac{(T_2 - T_1)}{\ln(r_2/r_1)} \frac{1}{r_1} = 2\pi \frac{k(T_1 - T_2)}{\ln(r_2/r_1)}$$

per unit length. The thermal resistance R is given by

$$R = \frac{\ln(r_2/r_1)}{2\pi k} \quad (2.32)$$

$$\dot{Q} = \frac{T_1 - T_2}{R}.$$

The cylindrical geometry can be viewed as a limiting case of the planar slab problem. To make the connection, consider the case when $\frac{r_2 - r_1}{r_1} \ll 1$. From the series expansion for $\ln(1+x)$ we recall that

$$\ln(1+x) \approx x - \frac{x^2}{2} + \frac{x^3}{3} + \dots \quad (2.33)$$

(Look it up, try it numerically, or use the binomial theorem on the series below and integrate term by term. $\frac{1}{1+x} = 1 - x + x^2 + \dots$)

The logarithms in Equation (2.31) can thus be written as

$$\ln\left(1 + \frac{r - r_1}{r_1}\right) \cong \frac{r - r_1}{r_1} \quad \text{and} \quad \ln \frac{r_2}{r_1} \cong \frac{r_2 - r_1}{r_1} \quad (2.34)$$

in the limit of $(r_2 - r_1) \ll r_1$. Using these expressions in equation (2.30) gives

$$T = (T_2 - T_1) \frac{(r - r_1)}{(r_2 - r_1)} + T_1. \quad (2.35)$$

With the substitution of $r - r_1 = x$, and $r_2 - r_1 = L$ we obtain

$$T = T_1 + (T_2 - T_1) \frac{x}{L} \quad (2.36)$$

which is the same as equation (2.17). The plane slab is thus the limiting case of the cylinder if $(r - r_1) / r \ll 1$, where the heat transfer can be regarded as taking place in (approximately) a planar slab. To see when this is appropriate, consider the expansion $\frac{\ln(1+x)}{x}$, which is the ratio of heat flux for a cylinder and a plane slab.

Table 2.2: Utility of plane slab approximation

x	.1	.2	.3	.4	.5
$\frac{\ln(1+x)}{x}$	.95	.91	.87	.84	.81

For < 10% error, the ratio of thickness to inner radius should be less than 0.2, and for 20% error, the thickness to inner radius should be less than 0.5.

A second example is the spherical shell with specified temperatures $T(r_1) = T_1$ and $T(r_2) = T_2$, as sketched in Figure 2.10.

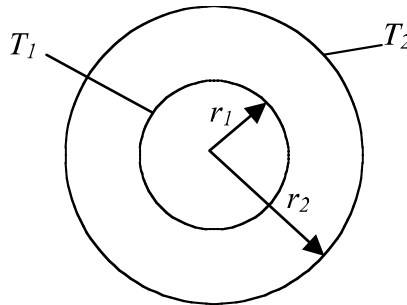


Figure 2.10: Spherical shell

The area is now $A(r) = 4\pi r^2$, so the equation for the temperature field is

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0. \quad (2.37)$$

Integrating equation (2.37) once yields

$$\frac{dT}{dr} = a / r^2. \quad (2.38)$$

Integrating again gives

$$T = -\frac{a}{r} + b$$

or, normalizing the spatial variable

$$T = \frac{a'}{(r/r_1)} + b \quad (2.39)$$

where a' and b are constants of integration. As before, we specify the temperatures at $r = r_1$ and $r = r_2$. Use of the first boundary condition gives $T(r_1) = T_1 = a' + b$. Applying the second boundary condition gives

$$T(r_2) = T_2 = \frac{a'}{(r_2/r_1)} + b$$

Solving for a' and b ,

$$\begin{aligned} a' &= \frac{T_1 - T_2}{1 - r_1/r_2} \\ b &= T_1 - \frac{T_1 - T_2}{1 - r_1/r_2}. \end{aligned} \quad (2.40)$$

In non-dimensional form the temperature distribution is thus:

$$\frac{T_1 - T}{T_1 - T_2} = \frac{1 - (r_1/r)}{1 - (r_1/r_2)} \quad (2.41)$$

Heat conduction through hollow composite pipe walls

Example: A steel pipe ($K = 45.0 \text{ W/m.K}$) having a 0.05m O.D is covered with a 0.042 m thick layer of magnesia ($K = 0.07\text{W/m.K}$) which in turn covered with a 0.024 m layer of fiberglass insulation ($K = 0.048 \text{ W/m.K}$). The pipe wall outside temperature is 370 K and the outer surface temperature of the fiberglass is 305K . What is the interfacial temperature between the magnesia and fiberglass? Also calculate the steady state heat transfer.

Given:

$\text{OD} = 0.05 \text{ m}$, $d_1 = 0.05 \text{ m}$, $r_1 = 0.025 \text{ m}$, $k_1 = 45 \text{ W/mK}$, $r_2 = r_1 + \text{thick of insulation 1}$
 $r_2 = 0.025 + 0.042$, $r_2 = 0.067 \text{ m}$, $k_2 = 0.07 \text{ W/mK}$, $k_3 = 0.048 \text{ W/mK}$,
 $r_3 = r_2 + \text{thick of insulation 2} = 0.067 + 0.024$, $r_3 = 0.091 \text{ m}$, $T_1 = 370 \text{ K}$, $T_3 = 305 \text{ K}$.

To find (i) T_2 and (ii) Q .

Solution :

Here thickness of pipe is not given; neglect the thermal resistance of pipe.

$$Q = \frac{(\Delta T)_{\text{overall}}}{\Sigma R_{th}}$$

Here $(\Delta T)_{\text{overall}} = T_1 - T_3 = 370 - 305 = 65 \text{ K}$

$$\Sigma R_{th} = R_{th1} + R_{th2}$$

$$R_{th1} = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_2 L} = \frac{\ln\left(\frac{0.067}{0.025}\right)}{2\pi \times 0.07 \times 1} = 2.2414 \text{ K/W}$$

$$R_{th2} = \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_3 L} = \frac{\ln\left(\frac{0.091}{0.067}\right)}{2\pi \times 0.048 \times 1} = 1.0152 \text{ K/W}$$

$$Q = \frac{65}{2.2414 + 1.0152} = 19.959 \text{ W/m}$$

To find T_2

$$Q = \frac{T_1 - T_2}{R_{th1}}$$

$$T_2 = T_1 - [Q \times R_{th1}]$$

$$= 370 - [19.959 \times 2.2414]$$

$$T_2 = 325.26 \text{ K}$$

Ref: <https://www.youtube.com/watch?v=R-5McNR2274>

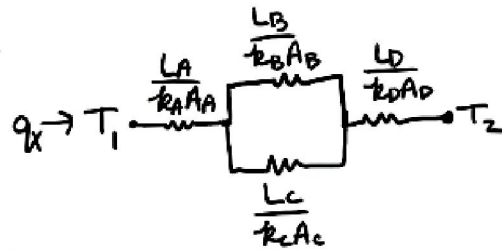
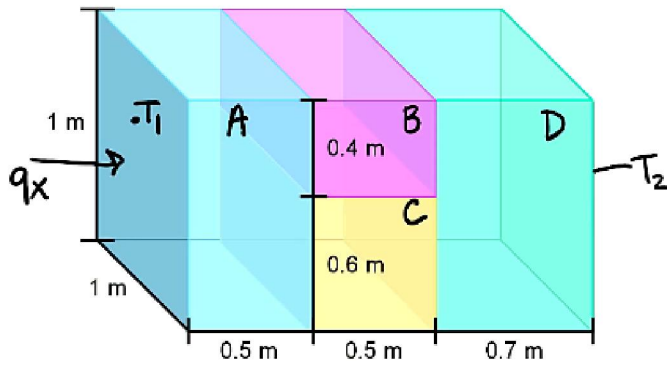
Composite wall with a series/parallel configuration

Given the information below, calculate the conductive heat transfer rate (q_x).

$$T_1 = 80^\circ\text{C}$$

$$T_2 = 25^\circ\text{C}$$

	$k \left(\frac{\text{W}}{\text{m}\cdot\text{K}} \right)$	$A \text{ (m}^2\text{)}$
A	0.038	1.0
B	0.12	0.4
C	1.7	0.6
D	0.04	1.0



$$\frac{L_A}{k_A A_A}, \frac{L_D}{k_D A_D}, \left[\frac{k_B A_B}{L_B} + \frac{k_C A_C}{L_C} \right]^{-1}$$

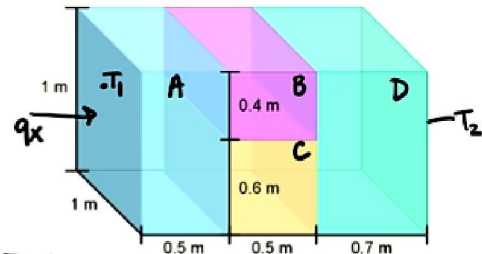
$$R_A = \frac{(0.5 \text{ m})}{(0.038 \frac{\text{W}}{\text{m}\cdot\text{K}})(1.0 \text{ m}^2)} = 13.2 \text{ K/W}$$

$$R_D = \frac{(0.7 \text{ m})}{(0.04 \frac{\text{W}}{\text{m}\cdot\text{K}})(1.0 \text{ m}^2)} = 17.5 \text{ K/W}$$

$$R_{B-C} = \left[\frac{(0.12 \frac{\text{W}}{\text{m}\cdot\text{K}})(0.4 \text{ m}^2)}{(0.5 \text{ m})} + \frac{(1.7 \frac{\text{W}}{\text{m}\cdot\text{K}})(0.6 \text{ m}^2)}{(0.5 \text{ m})} \right]^{-1}$$

$$R_{B-C} = 0.47 \text{ K/W}$$

$$q_x = \frac{\Delta T}{\sum R} = \frac{55 \text{ K}}{(13.2 + 17.5 + 0.47) \text{ K/W}} = \boxed{1.76 \text{ W}}$$



	$k \left(\frac{\text{W}}{\text{m}\cdot\text{K}} \right)$	$A \text{ (m}^2\text{)}$
A	0.038	1.0
B	0.12	0.4
C	1.7	0.6
D	0.04	1.0

Example 1.

The inside surface of a 25cm thick brick wall ($K = 0.69 \text{ W/mK}$) is at 27°C while the outside surface is at 2°C . Determine the rate of heat loss per unit area through the brick wall.

Soln: $Q = \frac{KA\Delta T}{L}$

as, $q = \frac{Q}{A} = \frac{K\Delta T}{L} = \frac{0.69 \times (27 - 2)}{0.25} = 69 \text{ W/m}^2$ **Ans.**

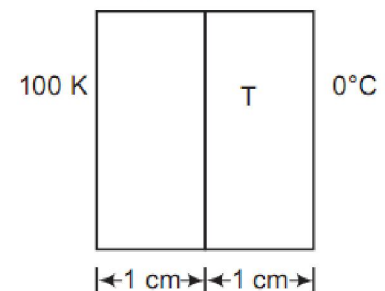
Example 2.

Sheets of brass and steel, each of thickness 1 cm, are placed in contact. The outer surface of brass is kept at 100°C and outer surface of steel is kept of 0°C . What is the temperature of the common interface? The thermal conductivities of brass and steel are in the ratio of 2 : 1.

Soln: $Q = K_b A \frac{100 - T}{L} = K_s A \frac{T - 0}{L}$

$$\frac{T}{100 - T} = \frac{K_b}{K_s} = 2$$

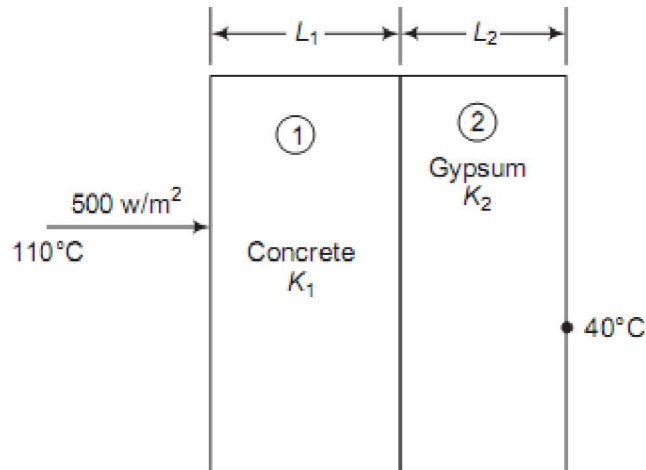
$$T = 66.7^\circ\text{C}$$

Ans.

Example 3.

A concrete wall ($K = 1.37 \text{ W/mk}$) of 10 cm thickness is to be plastered with gypsum ($K = 0.48 \text{ W/mk}$), so that the heat losses from the wall do not exceed 500 W/mk when the inner and outer surfaces of the wall are at 110°C and 40°C , respectively. Determine the thickness of plastering to be added to the concrete wall.

$$A = 1 \text{ m}^2$$



Soln:
$$R_1 = \frac{L_1}{AK_1} = \frac{0.1}{1.37} = 0.073 \text{ K/W}$$

$$R_2 = \frac{L_2}{AK_2} = \frac{L_2}{0.48} = 2.084 L_2 \text{ K/W}$$

$$Q = \frac{\Delta T}{R}$$

$$500 = \frac{110 - 40}{R}$$

$$R = 0.14 \text{ K/W}$$

Therefore,

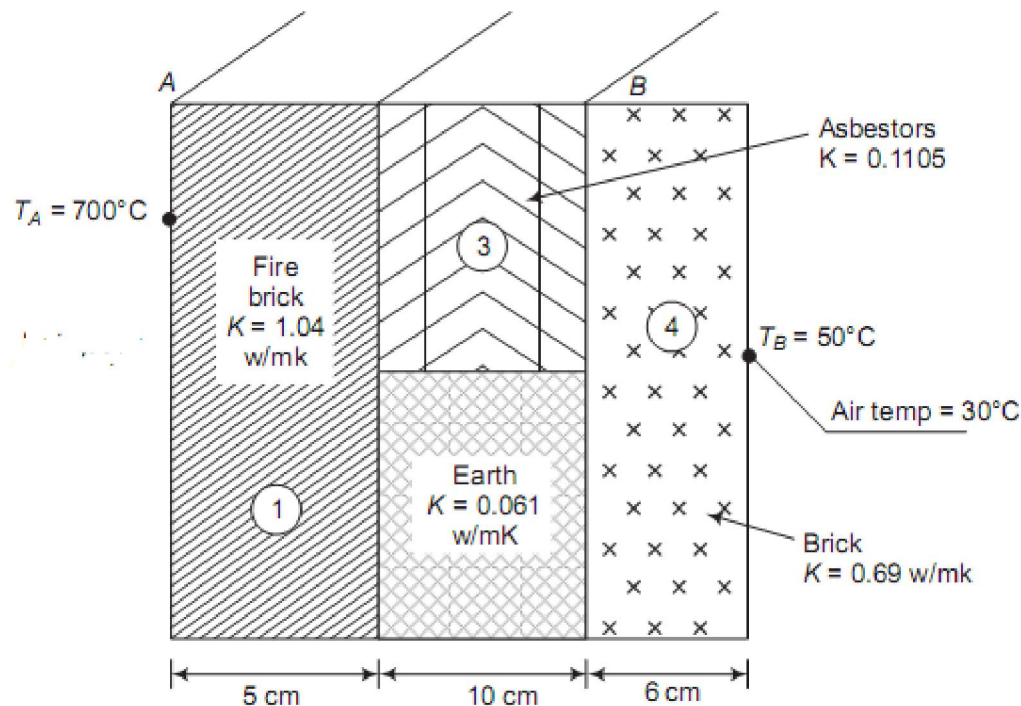
$$R = R_1 + R_2$$

$$0.14 = 0.073 + 2.084 L_2$$

$$L_2 = 0.032 \text{ m}$$

Ans.

Example 4. Find the rate of heat transfer per meter length of the compost wall.



$$K_1 = 1.04 \text{ W/mK}; K_2 = 0.061 \text{ W/mK};$$

$$K_3 = 0.1105 \text{ W/mK}; K_4 = 0.69 \text{ W/mK}$$

Soln: Let $A = 1 \text{ m}^2$

$$R_1 = \frac{l_1}{K_1 A_1} = \frac{0.05}{1.04 \times 0.25 \times 1} = 0.1923 \text{ K/W}$$

$$R_2 = \frac{l_2}{K_2 A_2} = \frac{0.10}{0.061 \times 0.1 \times 1} = 16.3934 \text{ K/W}$$

$$R_3 = \frac{l_3}{K_3 A_3} = \frac{0.10}{0.1105 \times 0.15 \times 1} = 60.332 \text{ K/W}$$

$$R_4 = \frac{l_4}{K_4 A_4} = \frac{0.05}{0.69 \times 0.25 \times 1} = 0.2899 \text{ K/W}$$

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{16.3934} + \frac{1}{60.332} = \frac{1}{4.4101}$$

or, $R_{23} = 4.4101 \text{ K/W}$

$$R = \sum R_i = R_1 + R_{23} + R_4 = 4.8923 \text{ K/W}$$

$$q = \frac{\Delta T}{R} = \frac{650}{4.8923} = 132.9 \text{ W}$$

Ans.

Example 5.

- The inner surface of a thick Invar ($K = 10.7 \text{ W / mK}$) cylinder of ID = 2 cm and OD = 12 cm is held at 150°C , while the outer surface is at 50°C . Calculate the temperature at $r = 2 \text{ cm}$, 3 cm , 4 cm and 5 cm .

Soln:
$$q_K = \frac{T_i - T_o}{\left\{ \frac{\ln(r_o / r_i)}{2\pi kl} \right\}} = \frac{150 - 50}{\left\{ \frac{\ln(6 / 1)}{2\pi \times 10.7 \times 1} \right\}} = 3.752 \text{ kW}$$

The temperature distribution is given by

$$T(r) = T_i - \frac{(T_i - T_o) \ln(r / r_i)}{\ln\left(\frac{r_o}{r_i}\right)} = T_i - \frac{(150 - 50)}{\ln(6)} \ln(r / 1)$$

$$T(r) = 150 - 55.81 \ln(r)$$

$$r = 2 \text{ cm}; T(r) = 111.3^\circ\text{C}; r = 3 \text{ cm}; T(r) = 88.7^\circ\text{C}$$

$$r = 4 \text{ cm}; T(r) = 72.6^\circ\text{C}; r = 5 \text{ cm}; T(r) = 60.2^\circ\text{C}$$